EXP 10	Performance Optimization and Best Practices in Zoho Creator Application Development

Aim:

To improve the performance of the Hotel Room Booking application by following best practices for forms, reports, workflows and Deluge scripts.

Theory:

Performance depends on how forms, reports, workflows and scripts are designed. Best practices used:

- **Lookup field** to link bookings with rooms instead of re-typing room details.
- **Auto Number** for Booking ID.
- **Report filters, sorting and grouping** so users see only needed records.
- **Single validation workflow** combining date, capacity and clash checks.
- **Conditional fetches** e.g. Room_Booking_Form[Room == input.Room && Status != "Checked-Out"].
- **Server-side calculation** of Nights and Total on successful submission.
- **Clear naming** of forms, fields and workflows.

Procedure:

1. Check that Lookup and Auto Number fields are used in the forms.
2. Apply filters, sorting and grouping in Room_Booking_Report.
3. Keep all validation logic in the single Validate Booking workflow.
4. Write Deluge queries that fetch only required records using conditions.
5. Run the recalculation function and test the application.

Deluge - recalculateAllBookings():

```
void recalculateAllBookings()
{
    // One-time sync of calculated fields for records entered before the workflows
existed
    for each b in Room_Booking_Form[ID != 0]
    {
        rm = Rooms[ID == b.Room];
        b.Room_Type = rm.Room_Type;
        b.Rate_Per_Night = rm.Rate_Per_Night;
        b.Nights = b.Check_In_Date.daysBetween(b.Check_Out_Date);
        b.Total_amount = b.Nights * rm.Rate_Per_Night;
        info "Booking " + b.Booking_ID + ": " + b.Nights + " nights x INR " +
rm.Rate_Per_Night + " = INR " + b.Total_amount;
    }
    for each rm in Rooms[ID != 0]
    {
        if(Room_Booking_Form[Room == rm.ID && Status == "Checked-In"].count() > 0)
        {
            rm.Room_Status = "Occupied";
        }
        else
        {
```

```
            rm.Room_Status = "Available";
        }
    }
}
```

Output:

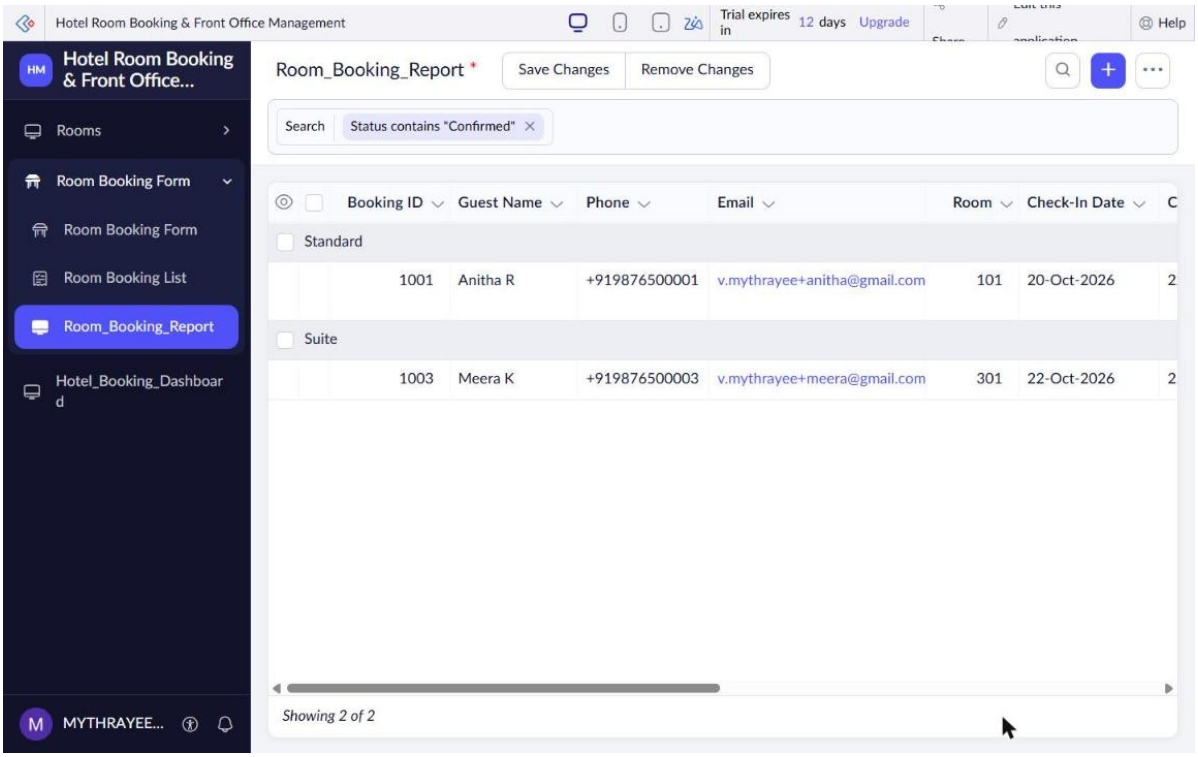


Fig. 10.1: Filtered report showing only required records

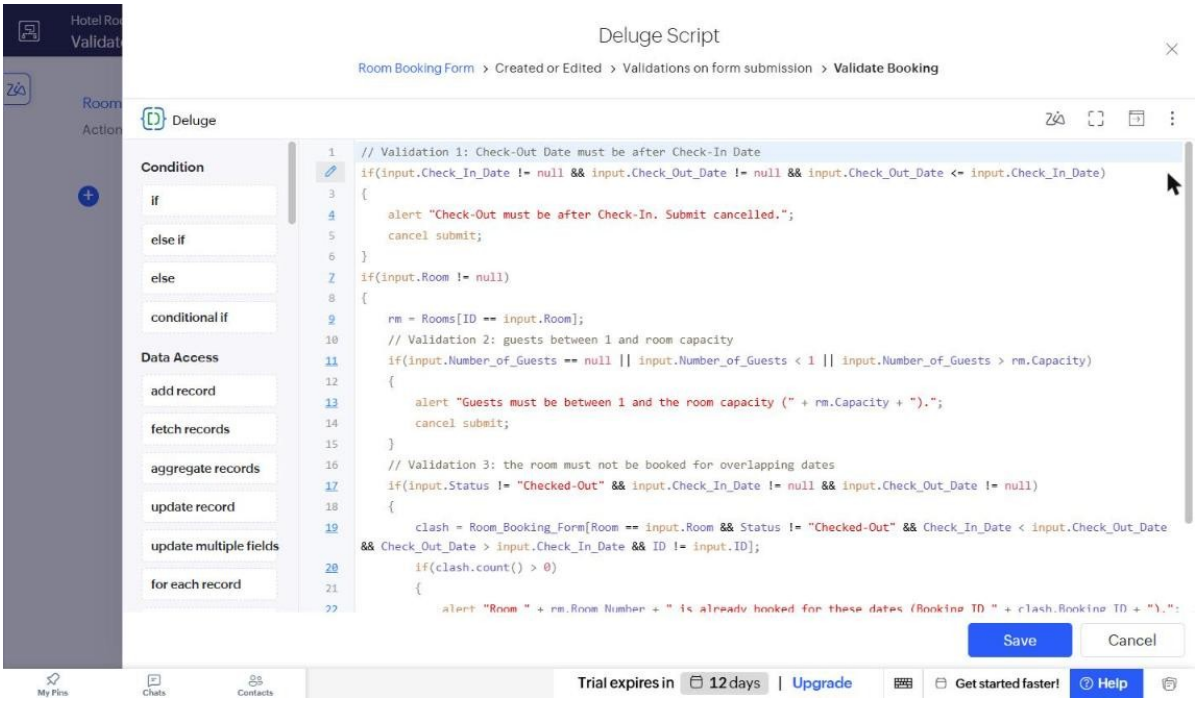


Fig. 10.2: Single validation workflow with conditional queries

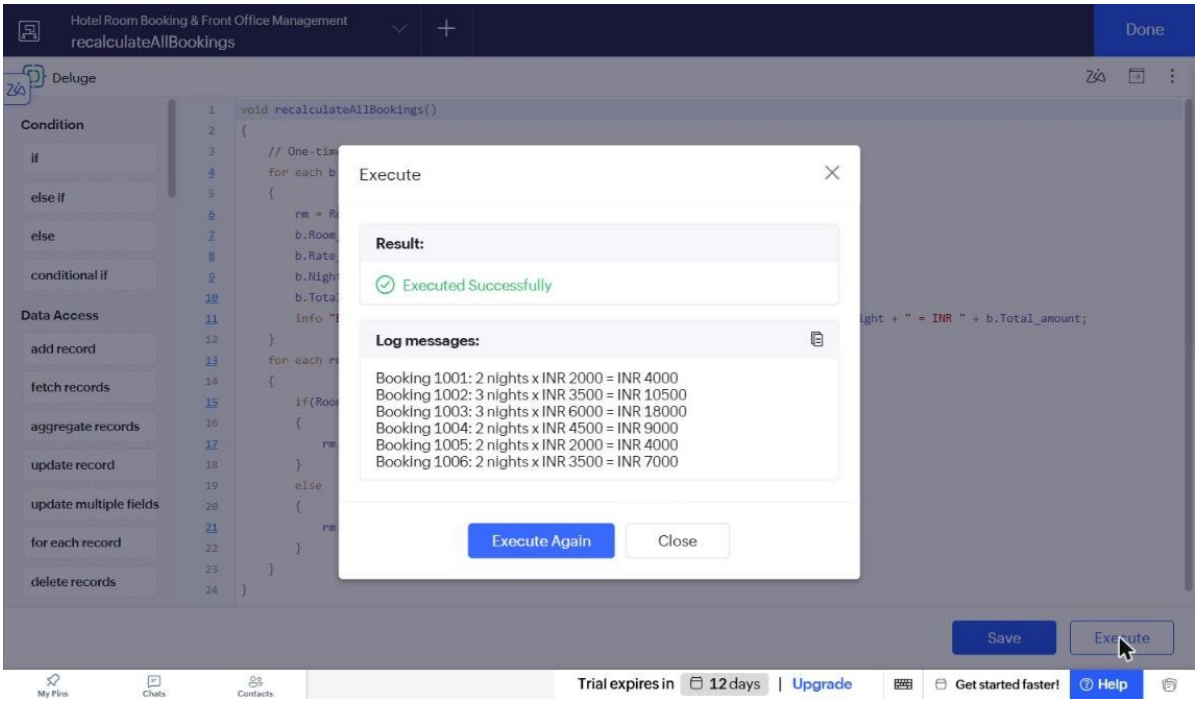


Fig. 10.3: Bill values calculated by the system for all bookings

Observation:

Practice	Benefit
Lookup + Auto Number	No duplicate data entry
One validation workflow	Fewer workflow executions
Conditional queries	Fetches only needed records
Report filters/grouping	Faster viewing of records

Result:

Best practices were applied to forms, reports, workflows and Deluge scripts, keeping the Hotel Room Booking application fast, consistent and easy to maintain.